
Unit – 2 : Session – 5 : SLO – 2 | Need for Threads and their Types — Scenarios

Scenario 1: gaming app (music, input, enemy AI in real time)

Thread Type: Kernel-Level Thread (KLT), so the threads can run truly in parallel on different cores.

Component: Program Counter, each thread keeps its own point of execution.

Need: Parallelism.

Scenario 2: Python threading with user-level scheduling

Thread Type: User-Level Thread (ULT), managed by the thread library without kernel help.

Component: Registers, saved and restored by the library on each thread switch.

Need: Responsiveness, other downloads keep going while one waits.

Scenario 3: kernel-mode threading, one blocked thread does not block the process

Thread Type: Kernel-Level Thread (KLT).

Component: Thread ID, the kernel tracks and schedules each thread separately.

Need: Responsiveness.

Scenario 4: own local variables, shared global variables

Thread Type: Kernel-Level Thread (threads of one server process).

Component: Stack, each thread has a private stack for its local variables.

Need: Resource Sharing, code and global data are shared inside the process.

Scenario 5: runtime maps user threads to kernel threads by load

Thread Type: Hybrid Model (many-to-many).

Component: Thread ID, used to map each user thread to a kernel thread.

Need: Scalability.